Discrete Electronics Project: Class AB Audio Amplifier Design and Simulation Report

María Fernanda Morales Calderón, Ian José Álvarez Arrieta,
Oscar Felipe Montejo Cordero, Bryan Alessandro Elizondo Fallas,
Andrés Obregón López
School of Electronics Engineering
Tecnológico de Costa Rica (TEC)
Analog One

***Abstract*—This paper presents the design and simulation of a discrete-component Class AB audio amplifier developed by Analog One. The system combines signal conditioning, voltage amplification, and a power stage to drive an $8\,\Omega$ loudspeaker. The design uses commercial component values and considers signal quality, transistor bias, and heat dissipation. The main calculations establish a $20\,\mathrm{W}$ output reference. In the recorded reference test, a $20\,\mathrm{V}$ peak input at $10\,\mathrm{kHz}$ produced $22.8\,\mathrm{W}$ of average load power. Calculated requirements and simulation readings are presented in separate tables to support a clear assessment of the design.**

***Index Terms*—Class AB, audio amplifier, discrete circuits, MOSFET, circuit simulation.**

I. Project Description

The project develops an audio amplifier that increases the level of an incoming audio signal and supplies the power needed by an $8\,\Omega$ loudspeaker. Developed by Analog One, the design aims to reproduce the signal at a useful output level while limiting distortion and power losses.

The amplifier contains three stages. The preamplifier conditions the incoming signal, the voltage amplifier increases its amplitude, and the power stage provides the current required by the loudspeaker. The Class AB output arrangement uses two complementary transistors that share the audio waveform. A small resting current helps smooth the transition between the positive and negative portions of the signal.

The design combines calculations, commercial component values, and circuit simulation. The evaluation considers output power, signal transfer, bias adjustment, and heat dissipation, together with the interaction between the stages.

II. Block Diagrams

A. Overall Amplifier

Figure 1 shows how the signal moves through the amplifier. Each stage has a specific role, from preparing the incoming signal to delivering power to the loudspeaker.

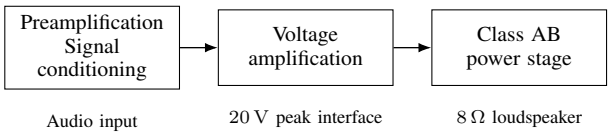


Figure 1. Main stages of the audio amplifier.

B. Preamplification Stage

Figure 2 shows the functional structure of the preamplification stage. The incoming signal passes through an AC-coupling network to remove any DC offset before reaching the discrete BJT amplifier. The single-ended Common-Emitter (CE) topology provides the necessary voltage amplification, while an unbypassed emitter resistor stabilizes the bias point and thermal drift. Finally, an output coupling network blocks the collector DC level, delivering a clean audio AC waveform centered at 0 V DC to the downstream mixing and power stages.

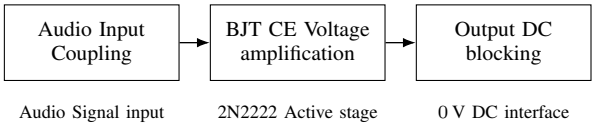


Figure 2. Functional blocks of the BJT preamplifier stage.

C. Voltage Amplification Stage

The voltage amplifier connects the preamplifier to the input of the power stage. Figure 3 follows the signal from left to right. Each IRF530 inverts the signal at its drain, so the two stages together preserve the input polarity near midband. The IRF9540 supplies current to the second stage.

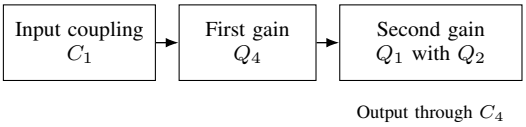


Figure 3. Signal path through the voltage amplification stage.

D. Class AB Power Stage

The Class AB arrangement was selected because it combines a smooth transition between the two portions of an audio signal with moderate resting power consumption. Its overlapping conduction helps reduce crossover distortion, which is particularly relevant when reproducing small changes in the waveform [1, sec. 7–2].

The selected circuit also provides the current needed by the loudspeaker while preserving the signal polarity and most of its voltage amplitude. These features make it suitable as the final stage of an amplifier whose preceding stages already provide the required voltage level [1, sec. 9–2].

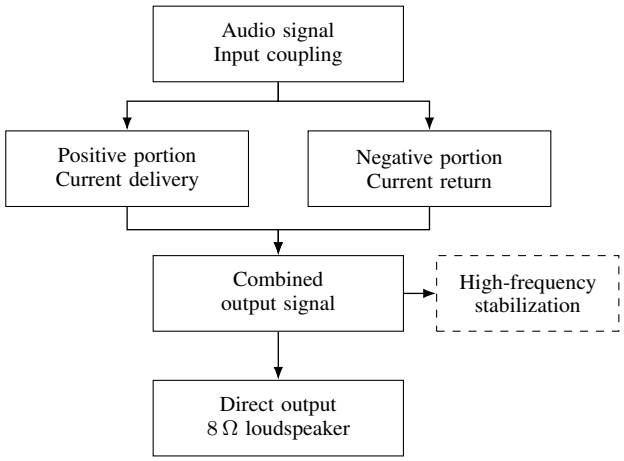


Figure 4. Functional blocks of the selected Class AB output arrangement.

III. Design Considerations and Calculated Values

A. Preamplification Stage

The preamplifier uses a 2N2222 NPN BJT in a Common-Emitter (CE) configuration powered by a single positive supply $V_{CC} = 24$ V (or $+12$ V single rail domain). Figure **??** presents the complete schematic diagram of the stage. The design uses an unbypassed emitter resistor R_4 to fix the AC voltage gain independently of transistor parameter variations (β) and to minimize thermal drift.

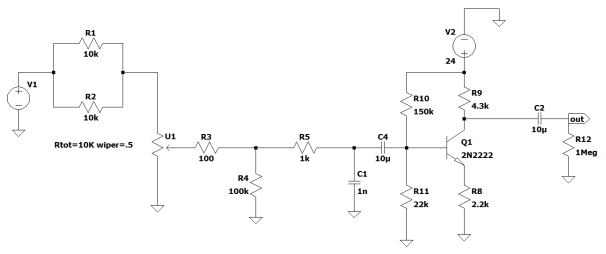


Figure 5. Preamplifier schematic stage featuring 2N2222 BJT.

1) DC Operating Point Analysis: The DC bias network is established by the base voltage divider R_1 and R_2, collector resistor R_3, and emitter resistor R_4. Coupling capacitors C_1 and C_2 act as open circuits for DC.

The base Thévenin equivalent resistance R_{Th} and voltage V_{Th} are:

$$R_{Th} = R_1 \parallel R_2 = \frac{(68\ \text{k}\Omega)(33\ \text{k}\Omega)}{68\ \text{k}\Omega + 33\ \text{k}\Omega} \approx 22.22\ \text{k}\Omega \quad (1)$$

$$V_{Th} = V_{CC}\left(\frac{R_2}{R_1 + R_2}\right) = 24\ \text{V}\left(\frac{33\ \text{k}\Omega}{68\ \text{k}\Omega + 33\ \text{k}\Omega}\right) \approx 7.84\ \text{V} \quad (2)$$

Assuming $V_{BE} = 0.7$ V and a nominal current gain $\beta = 150$, the base bias current I_{BQ} is:

$$I_{BQ} = \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_4} = \frac{7.84\ \text{V} - 0.7\ \text{V}}{22.22\ \text{k}\Omega + (151)(2.2\ \text{k}\Omega)} \approx 20.14\ \mu\text{A} \quad (3)$$

The quiescent collector current I_{CQ} and emitter current I_{EQ} are:

$$I_{CQ} = \beta I_{BQ} = (150)(20.14\ \mu\text{A}) \approx 3.02\ \text{mA} \quad (4)$$

$$I_{EQ} = (\beta + 1)I_{BQ} \approx 3.04\ \text{mA} \quad (5)$$

The resulting DC voltages at the emitter (V_E), collector (V_C), and collector-to-emitter (V_{CEQ}) are calculated as:

$$V_E = I_{EQ}R_4 = (3.04\ \text{mA})(2.2\ \text{k}\Omega) \approx 6.69\ \text{V} \quad (6)$$

$$V_C = V_{CC} - I_{CQ}R_3 = 24\ \text{V} - (3.02\ \text{mA})(4.7\ \text{k}\Omega) \approx 9.81\ \text{V} \quad (7)$$

$$V_{CEQ} = V_C - V_E = 9.81\ \text{V} - 6.69\ \text{V} = 3.12\ \text{V} \quad (8)$$

The Q-point $(V_{CEQ}, I_{CQ}) = (3.12\ \text{V}, 3.02\ \text{mA})$ ensures stable operation within the linear active region.

2) Small-Signal AC Voltage Gain: The dynamic emitter resistance r_e at room temperature ($V_T \approx 25.85$ mV) is:

$$r_re = \frac{V_T}{I_{CQ}} = \frac{25.85\ \text{mV}}{3.02\ \text{mA}} \approx 8.56\ \Omega \quad (9)$$

Because the emitter resistor R_4 is fully unbypassed to control gain and prevent distortion/clipping, the small-signal voltage gain A_v of the stage is given by:

$$A_v = \frac{v_{out}}{v_{in}} \approx -\frac{R_3 \parallel R_L}{r_e + R_4} \approx -\frac{4.7\ \text{k}\Omega}{8.56\ \Omega + 2.2\ \text{k}\Omega} \approx -2.13\ \text{V/V} \quad (10)$$

3) Thermal and Power Dissipation Analysis: At ambient temperature $T_A = 25°\text{C}$, the quiescent power dissipation of the transistor Q_1 is:

$$P_{D,Q1} = V_{CEQ}I_{CQ} = (3.12\ \text{V})(3.02\ \text{mA}) \approx 9.42\ \text{mW} \quad (11)$$

Using the junction-to-ambient thermal resistance $\theta_{JA} = 200°\text{C/W}$ for the TO-92 package, the operating junction temperature T_J is:

$$T_J = T_A + P_{D,Q1}\theta_{JA} = 25°\text{C} + (0.00942\ \text{W})(200°\text{C/W}) \approx 26.88°\text{C} \quad (12)$$

This negligible temperature rise ($\Delta T \approx 1.88°\text{C}$) confirms that no external heat sink is required.

Table I
CALCULATED PREAMPLIFIER BIAS AND PERFORMANCE PARAMETERS

Parameter	Calculated Value
Base Voltage (V_B)	7.39 V
Emitter Voltage (V_E)	6.69 V
Collector Voltage (V_C)	9.81 V
Quiescent Current (I_{CQ})	3.02 mA
Collector-Emitter Voltage (V_{CEQ})	3.12 V
Stage AC Voltage Gain (A_v)	−2.13 V/V
Transistor Power Dissipation ($P_{D,Q1}$)	9.42 mW
Estimated Junction Temp. (T_J)	26.88°C

B. Voltage Amplification Stage

Figure 6 shows the Multisim circuit used for the final test. The design target was a 20 V peak sine wave at the power-stage input from a 2 V RMS signal, using ±24 V supplies. The corresponding output voltage and required gain are

$$V_{\text{out,rms}} = 20/\sqrt{2} = 14.14\,\text{V}, \tag{13}$$

$$A_{v,\text{target}} = 14.14/2 = 7.07\,\text{V/V}. \tag{14}$$

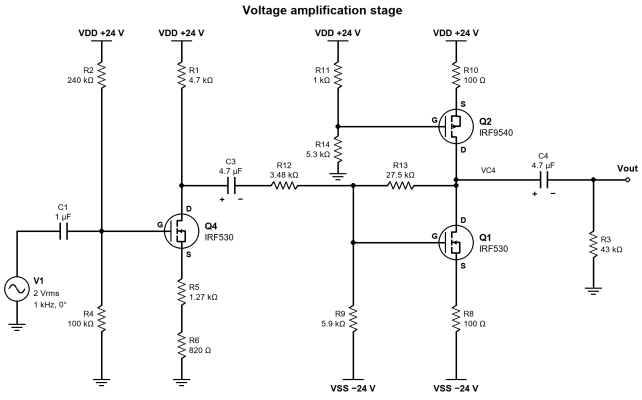


Figure 6. Voltage amplification stage as tested in Multisim. VC4 is the common drain node before the output capacitor. The 43 kΩ resistor represents the load used for this stage test.

1) DC operating point: At zero input, C_1 isolates the gate of Q_4 from the source. Neglecting gate current, its bias divider gives

$$V_{G4} = 24\,\text{V}\frac{100}{240+100} = 7.06\,\text{V}, \tag{15}$$

$$R_{S4} = R_5 + R_6 = 2.09\,\text{k}\Omega. \tag{16}$$

Using the simulated Q_4 drain level of 16.89 V and the 4.7 kΩ drain resistor, the first-stage bias can be reconstructed without assuming a fixed MOSFET threshold:

$$I_{D4} = (24 - 16.89)\,\text{V}/R_1 = 1.51\,\text{mA}, \tag{17}$$

$$V_{S4} = I_{D4}R_{S4} = 3.16\,\text{V}, \tag{18}$$

$$V_{GS4} = 7.06 - 3.16 = 3.90\,\text{V}, \tag{19}$$

$$V_{DS4} = 16.89 - 3.16 = 13.73\,\text{V}. \tag{20}$$

For Q_2, the 1.0 kΩ and 5.3 kΩ divider sets the gate voltage to $24(5.3/6.3) = 20.19$ V. In the second stage, R_{13} feeds the drain voltage back to the gate of Q_1, and R_9 returns the gate to −24 V. With the simulated resting voltage $V_X = -0.125$ V at VC4, and negligible gate current, the DC path gives

$$I_{\text{fb}} = (V_X + 24\,\text{V})/(R_{13} + R_9) = 0.715\,\text{mA}, \tag{21}$$

$$V_{G1} = -24\,\text{V} + I_{\text{fb}}R_9 = -19.78\,\text{V}. \tag{22}$$

This places the internal output close to the midpoint of the supplies. A 20 V excursion about that point leaves approximately 4 V to each rail, before allowing for voltage drops in the devices and source resistors.

2) Signal coupling and power: The input resistance set by the gate divider is $R_2 \parallel R_4 = 70.6\,\text{k}\Omega$. With the ideal test source, the input coupling corner is $1/(2\pi C_1(R_2 \parallel R_4)) = 2.25$ Hz. At the output, C_4 blocks the internal DC level while R_3 returns Vout to ground. At the target amplitude the load draws $14.14\,\text{V}/43\,\text{k}\Omega = 0.329$ mA RMS and dissipates $14.14^2/43000 = 4.65$ mW.

At rest, Q_4 dissipates approximately $V_{DS4}I_{D4} = 20.8$ mW. The Q_2 gate divider draws $24\,\text{V}/(6.3\,\text{k}\Omega) = 3.81$ mA. Its resistor losses, calculated as I^2R, are 14.5 mW in R_{11} and 76.9 mW in R_{14}.

For an estimate of the second-stage resting losses, take $V_{GS1} \approx 3.91$ V, based on the observed gate-to-source voltage. Then $V_{S1} \approx -23.69$ V and $I_{D1} \approx (V_{S1} + 24\,\text{V})/R_8 = 3.07$ mA. The active load supplies both this current and the feedback current, giving $I_{Q2} \approx 3.79$ mA and $V_{S2} \approx 24\,\text{V} - I_{Q2}R_{10} = 23.62$ V. Multiplying each drain-to-source voltage magnitude by its estimated current gives approximately 72 mW for Q_1 and 90 mW for Q_2. These are nominal bias estimates, distinct from the losses under signal drive.

C. Class AB Power Stage

Figure 7 presents the power-stage circuit. The design retains the complementary output arrangement, with nominal supplies of +24 V and −24 V. The load is connected directly to VOUT, the junction between RS1 and RS2.

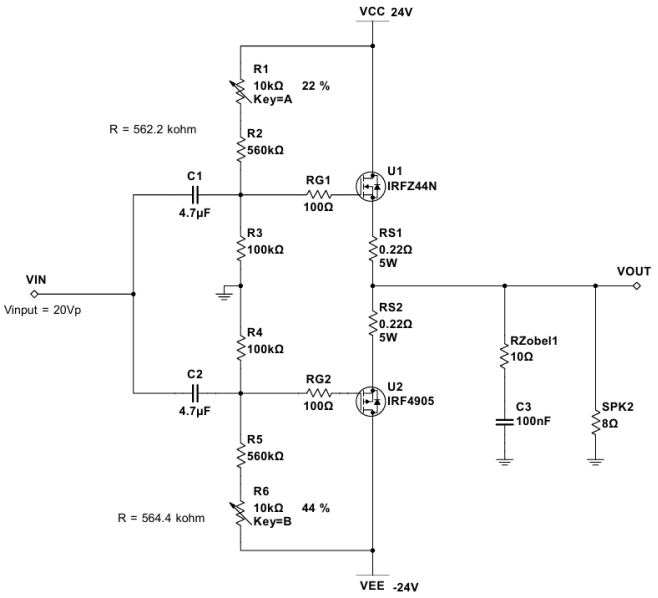


Figure 7. Class AB power-stage schematic with IRFZ44N and IRF4905 MOSFETs.

1) Output Power Requirement: The design reference is 20 W delivered to an $8\,\Omega$ load. For a sinusoidal signal, the average power is

$$P_L = \frac{V_{L,\mathrm{rms}}^2}{R_L}. \tag{23}$$

Here, RMS is the effective voltage value used to calculate average audio power. The required load voltage and current are

$$V_{LOAD,\mathrm{rms}} = \sqrt{(20)(8)} \approx 12.65\,\mathrm{V}, \tag{24}$$

$$V_{LOAD,peak} = \sqrt{2}(12.65) \approx 17.89\,\mathrm{V}, \tag{25}$$

$$I_{LOAD,\mathrm{rms}} = \frac{12.65}{8} \approx 1.58\,\mathrm{A}. \tag{26}$$

These values define the load requirements. With the specified 20 V peak input, the output power depends on the voltage transfer of the stage. Table II summarizes the calculated values.

Table II
CALCULATED OUTPUT REQUIREMENTS FOR THE 20 W REFERENCE

Quantity	Calculated value
Average output power, reference	20 W
Load RMS voltage	12.65 V
Load peak voltage	17.89 V
Load peak-to-peak voltage	35.78 V
Load RMS current	1.58 A
Load peak current	2.24 A

2) Class AB Bias Adjustment: Bias sets the transistor operating conditions before an audio signal is applied. The design adopts a quiescent drain-current target of $I_{DQ} = 50\,\mathrm{mA}$ in each branch. This small resting current provides the intended overlap between the two transistors near the signal transition [1, sec. 7–2].

The $0.22\,\Omega$ source resistors provide a convenient current reference. At the selected target, the drop across each resistor is

$$V_{RS,Q} = I_{DQ}R_S = (0.050)(0.22) = 11\,\mathrm{mV}. \tag{27}$$

Bias adjustment targets these drops while keeping VOUT close to 0 V DC. Because SPK2 is connected directly to this node, the same DC voltage appears across the load.

The updated component labels give the effective bias resistances

$$R_N = R2 + R1_{\mathrm{eff}}, \tag{28}$$

$$R_P = R5 + R6_{\mathrm{eff}}. \tag{29}$$

The nominal gate voltages follow the two dividers:

$$V_{G,U1} = 24\,\mathrm{V}\frac{R3}{R_N + R3}, \tag{30}$$

$$V_{G,U2} = -24\,\mathrm{V}\frac{R4}{R_P + R4}. \tag{31}$$

With $R3 = R4 = 100\,\mathrm{k}\Omega$, the initial gate references of $+3.6243\,\mathrm{V}$ and $-3.6123\,\mathrm{V}$ correspond to $R_N = 562.2\,\mathrm{k}\Omega$ and $R_P = 564.4\,\mathrm{k}\Omega$. Commercial $560\,\mathrm{k}\Omega$ resistors R2 and R5 are combined with $10\,\mathrm{k}\Omega$ trimmers R1 and R6. Their nominal effective settings are $2.2\,\mathrm{k}\Omega$ and $4.4\,\mathrm{k}\Omega$, respectively, or 22% and 44% of the trimmer range.

These divider settings provide the initial gate voltages. The $50\,\mathrm{mA}$ target is assessed through the separate DC operating-point analysis and the drops in (27).

Table III
CALCULATED BIAS REFERENCES AND ADJUSTMENT TARGETS

Quantity	Design reference
U1 gate bias, initial	$+3.6243\,\mathrm{V}$
U2 gate bias, initial	$-3.6123\,\mathrm{V}$
$R_N = R2 + R1_{\mathrm{eff}}$	$562.2\,\mathrm{k}\Omega$
$R_P = R5 + R6_{\mathrm{eff}}$	$564.4\,\mathrm{k}\Omega$
R1 effective setting	$2.2\,\mathrm{k}\Omega$
R6 effective setting	$4.4\,\mathrm{k}\Omega$
I_{DQ}, target per MOSFET	$50\,\mathrm{mA}$
RS1 and RS2 drop at target	$11\,\mathrm{mV}$ each
VOUT at rest, target	0 V DC

3) Signal Coupling: C1 and C2 transfer the audio signal to the gates while keeping their DC bias settings separate from the preceding stage. Each capacitor is $4.7\,\mu\mathrm{F}$. With a low-impedance driving source, the divider resistance seen by each capacitor is approximately $84.9\,\mathrm{k}\Omega$, giving an input coupling corner near $0.40\,\mathrm{Hz}$. C1 and C2 are polarized electrolytics. With zero DC at VIN, the positive terminal of C1 faces the upper bias node, while the positive terminal of C2 faces VIN.

The output is connected directly to the $8\,\Omega$ load. RZobel1 and C3 form the series output RC network, with values of $10\,\Omega$ and $100\,\mathrm{nF}$. This branch provides a high-frequency load in parallel with the loudspeaker.

4) Heat Dissipation: All thermal calculations use an ambient temperature of $T_A = 25\,^\circ\mathrm{C}$, both at rest and during signal operation. The analysis relates each component's power loss to its estimated steady-state temperature. Table IV summarizes the results and the thermal-resistance parameters used in the calculation.

Power losses: At $I_{DQ} = 50\,\mathrm{mA}$, the resting losses in each MOSFET and source resistor are

$$P_{D,Q} = (24 - I_{DQ}R_S)I_{DQ} \simeq 1.20\,\mathrm{W}, \tag{32}$$

$$P_{RS,Q} = I_{DQ}^2 R_S = 0.55\,\mathrm{mW}. \tag{33}$$

Resistor losses follow $P_R = I_{\mathrm{rms}}^2 R$. For example, the recorded gate RMS voltage gives

$$P_{R3} = P_{R4} = \frac{14.6^2}{100000} = 2.13\,\mathrm{mW}. \tag{34}$$

The same calculation gives $1.09\,\mathrm{mW}$ in R2 and $1.08\,\mathrm{mW}$ in R5. R1 and R6 dissipate $4.28\,\mu\mathrm{W}$ and $8.50\,\mu\mathrm{W}$, respectively.

For the gate-resistor loss calculation, the estimated RMS currents are $1.306\,\mathrm{mA}$ and $3.022\,\mathrm{mA}$:

$$P_{RG1} = (1.306 \times 10^{-3})^2(100) \simeq 0.171\,\mathrm{mW}, \tag{35}$$

$$P_{RG2} = (3.022 \times 10^{-3})^2(100) \simeq 0.913\,\mathrm{mW}. \tag{36}$$

Capacitor losses follow $P_C = I_{C,\mathrm{rms}}^2 r_{ESR}$, where ESR represents the capacitor's series loss resistance. Using a fixed ESR of $34\,\Omega$ and estimated currents of $1.317\,\mathrm{mA}$ and $3.026\,\mathrm{mA}$ gives

$$P_{C1} = (1.317 \times 10^{-3})^2(34) \simeq 0.0590\,\mathrm{mW}, \tag{37}$$

$$P_{C2} = (3.026 \times 10^{-3})^2(34) \simeq 0.311\,\mathrm{mW}. \tag{38}$$

These gate and input-capacitor currents are calculated with a constant input-capacitance approximation for the MOSFETs.

At $10\,\mathrm{kHz}$, the C3 loss calculation uses $\tan\delta = 0.025$:

$$X_{C3} = \frac{1}{2\pi f C3} = 159.15\,\Omega, \tag{39}$$

$$r_{ESR,3} = \tan\delta\, X_{C3} = 3.98\,\Omega. \tag{40}$$

The Zobel-branch current is

$$I_Z = \frac{13.5}{\sqrt{(10 + 3.98)^2 + 159.15^2}} \simeq 84.50\,\mathrm{mA}. \tag{41}$$

Therefore,

$$P_{\mathrm{RZobel1}} = I_Z^2(10) \simeq 0.0714\,\mathrm{W}, \tag{42}$$

$$P_{C3} = I_Z^2 r_{ESR,3} \simeq 0.0284\,\mathrm{W}. \tag{43}$$

The source-resistor sizing calculation includes the load current, the Zobel-branch current, and I_{DQ}:

$$P_{RS,\mathrm{design}} = 0.22(1.69 + 0.0845 + 0.050)^2 \simeq 0.732\,\mathrm{W}. \tag{44}$$

This power allowance applies to each $5\,\mathrm{W}$ source resistor.

Component temperatures: The passive-component temperature calculation is

$$T_i = T_A + R_{\theta i} P_i. \tag{45}$$

For example,

$$T_{RS1} = T_{RS2} \simeq 25 + (0.732)(55) = 65.3\,^\circ\mathrm{C}, \tag{46}$$

$$T_{C3} \simeq 25 + (0.0284)(1157) = 57.9\,^\circ\mathrm{C}. \tag{47}$$

The same expression is applied to the remaining passive components using the power and thermal resistance listed in Table IV.

For the MOSFETs, the calculation includes a separate $4\,^\circ\mathrm{C/W}$ heat sink for each device and a $0.5\,^\circ\mathrm{C/W}$ mounting interface:

$$T_J = T_A + P_D(R_{\theta JC} + R_{\theta CS} + R_{\theta SA}). \tag{48}$$

The datasheet junction-to-case thermal resistances are $1.5\,^\circ\mathrm{C/W}$ for U1 and $0.75\,^\circ\mathrm{C/W}$ for U2 [2], [3]. Using the recorded transistor losses gives

$$T_{J,U1} = 25 + 6.58(1.5 + 0.5 + 4) \simeq 64.5\,^\circ\mathrm{C}, \tag{49}$$

$$T_{J,U2} = 25 + 6.63(0.75 + 0.5 + 4) \simeq 59.8\,^\circ\mathrm{C}. \tag{50}$$

At rest, the calculated junction temperatures are $32.2\,^\circ\mathrm{C}$ for U1 and $31.3\,^\circ\mathrm{C}$ for U2. At the $10\,\mathrm{W}$ per-device design allowance, they are $85\,^\circ\mathrm{C}$ and $77.5\,^\circ\mathrm{C}$, below the selected $100\,^\circ\mathrm{C}$ design limit.

Table IV
CALCULATED COMPONENT TEMPERATURES AT 25 °C

Component	Power used	R_θ (°C/W)	Temperature (°C)
U1	6.58 W	6.00	64.5
U2	6.63 W	5.25	59.8
R1	4.28 μW	263	25.0
R2	1.09 mW	140	25.2
R3, R4	2.13 mW each	140	25.3 each
R5	1.08 mW	140	25.2
R6	8.50 μW	263	25.0
RG1	0.171 mW	140	25.0
RG2	0.913 mW	140	25.1
RS1, RS2	0.732 W each	≃ 55	65.3 each
RZobel1	0.0714 W	75	30.4
C1	0.0590 mW	472	25.0
C2	0.311 mW	472	25.1
C3	0.0284 W	1157	57.9

Temperatures are steady-state analytical estimates at 25 °C, rounded to 0.1 °C. U1/U2 entries are junction temperatures; passive entries are body temperatures. The R_θ column lists the calculation parameters. RS1/RS2 use a power allowance for component sizing. Gate and capacitor losses are analytical estimates.

IV. COMPONENT SELECTION CRITERIA

A. Preamplification Stage

The selection of passive and active components for the preamplifier focuses on signal integrity, gain stability, and minimal noise introduction. Table V summarizes the chosen components and their design specifications.

- **Transistor (Q_1):** The 2N2222 general-purpose NPN transistor was chosen due to its wide linear operational range, high gain-bandwidth product ($f_T > 250$ MHz), and ready availability in TO-92 packages.
- **Emitter Resistor (R_4):** Left unbypassed to introduce negative local feedback. This lowers non-linear distortion

Table V
PREAMPLIFIER COMPONENTS AND SELECTION RATINGS

Reference	Nominal Value	Selection Rating / Purpose
Q_1	2N2222	NPN BJT, general-purpose switching/amp
R_1	68 kΩ	Metal film, 1%, 0.25 W (Upper Bias)
R_2	33 kΩ	Metal film, 1%, 0.25 W (Lower Bias)
R_3	4.7 kΩ	Metal film, 1%, 0.25 W (Collector Load)
R_4	2.2 kΩ	Metal film, 1%, 0.25 W (Feedback Emitter)
R_5	100 kΩ	Metal film, 1%, 0.25 W (Output Pull-down)
C_1	470 nF	Film/Ceramic, 50 V (Input AC Coupling)
C_2	2.2 μF	Electrolytic/Film, 50 V (Output DC Block)
C_3	1 nF	Ceramic, 50 V (High-Freq Noise Filtering)

and prevents gain drift caused by temperature or β variations.

- **Coupling Capacitors (C_1, C_2):** Sized to maintain a lower cutoff corner frequency well below the audible spectrum ($f_L < 20$ Hz), preventing baseline attenuation of low-frequency audio content.
- **Output Pull-down (R_5):** Provides a continuous reference to ground at the output side of C_2, guaranteeing a zero DC baseline before entering the operational amplifier mixing stage.

B. Voltage Amplification Stage

Two IRF530 N-channel MOSFETs [4] were selected for voltage gain and an IRF9540 P-channel MOSFET [5] for the active load. Its maximum drain-to-source voltage is -100 V. Both models were available in the Multisim library and listed by the local supplier reviewed for the project. The source resistors provide local feedback. The values of the input and feedback resistors were tested together because they affect both the gain of Q_1 and the load seen by Q_4.

Table VI
VOLTAGE-STAGE COMPONENT VALUES USED IN THE FINAL SIMULATION.

Reference	Value	Function
Q_4, Q_1	IRF530	Common-source gain
Q_2	IRF9540	Active load
R_2, R_4	240, 100 kΩ	Q_4 gate bias
R_1	4.7 kΩ	Q_4 drain load
R_5, R_6	1.27 kΩ, 820 Ω	Q_4 source path
R_{12}, R_{13}	3.48, 27.5 kΩ	Input and feedback
R_9	5.9 kΩ	Return to -24 V
R_8, R_{10}	100 Ω each	Source resistors
R_{11}, R_{14}	1, 5.3 kΩ	Q_2 gate bias
R_3	43 kΩ	Output test load
C_1	1 μF	Nonpolar input coupling
C_3, C_4	4.7 μF each	Interstage and output

C. Class AB Power Stage

The selected components retain the circuit topology and use commercial nominal values. Table VII lists the main components and the ratings considered in their selection.

Table VII
POWER-STAGE COMPONENTS AND SELECTION RATINGS

Reference	Value or model	Selection rating
U1	IRFZ44N	N-channel MOSFET
U2	IRF4905	P-channel MOSFET
R2, R5	560 kΩ	1%, 0.25 W
R3, R4	100 kΩ	1%, 0.25 W
R1, R6	10 kΩ	Trimmers, 0.25 W
RG1, RG2	100 Ω	0.25 W
RS1, RS2	0.22 Ω	5 W
C1, C2	4.7 μF	50 V, electrolytic
RZobel1	10 Ω	2 W
C3	100 nF	50 V, ceramic, code 104
SPK2	8 Ω	30 W nominal rating

The MOSFET voltage ratings are considered together with the ± 24 V supplies and the expected signal excursions [2], [3]. The source-resistor power ratings provide margin for the load current, while the gate resistors and output RC network help control high-frequency behavior.

The heat-sink selection follows the stated dissipation allowance. The component references in Table VII match the updated schematic, including C1/C2 as electrolytic coupling capacitors and C3 as the 100 nF ceramic Zobel capacitor marked 104.

V. SIMULATION METHODOLOGY

A. Common Simulation Conditions

Multisim provides the circuit environment for the reported test. The power-stage input is a 20 V peak sinusoidal signal at 10 kHz, with zero DC offset. The circuit uses nominal ± 24 V supplies and an 8 Ω load. R1 and R6 have nominal settings of 22% and 44%, respectively. These settings establish the initial divider voltages; the DC analysis assesses I_{DQ} separately.

B. Preamplification Stage

The simulation of the BJT preamplifier stage was conducted in both LTspice and NI Multisim to verify DC biasing, dynamic transient behavior, and overall voltage amplification before physical assembly.

1) **DC Operating Point Analysis:** The simulation evaluates the circuit under zero-input signal condition ($v_{in} = 0$ V AC). It measures the base voltage (V_B), emitter voltage (V_E), collector voltage (V_C), and collector bias current (I_{CQ}) of transistor Q_1. The objective is to verify that the operating point resides in the linear active region ($V_{CEQ} \approx 3.12$ V, $I_{CQ} \approx 3.02$ mA), avoiding saturation or cutoff limits and providing adequate headroom for signal swings.
2) **Transient Analysis:** A 1 kHz sinusoidal voltage source (V_1) was applied at the input coupling capacitor C_1. Transient simulations were run with a time step sufficient to capture multiple complete audio cycles (e.g., $t_{stop} =$ 10 ms). The transient response measures the peak-to-peak input and output voltage waveforms, evaluating the stage voltage gain $A_v = v_{out}/v_{in}$ and checking for

asymmetrical clipping or waveform distortion caused by high signal amplitudes.

3) **Frequency Response and Filtering:** An AC sweep simulation (10 Hz to 100 kHz) was executed to determine the lower and upper -3 dB cutoff frequencies (f_L and f_H). This analysis assesses the high-pass response determined by coupling capacitors C_1 and C_2, as well as the high-frequency attenuation introduced by filtering capacitor C_3.

C. Voltage Amplification Stage

The voltage stage was tested separately in Multisim using the $43\,\mathrm{k}\Omega$ load in Fig. 6. First, the source amplitude was set to zero and the voltage at VC4 was recorded. Next, a $2\,\mathrm{V}$ RMS, $1\,\mathrm{kHz}$ sine wave was applied. Cursors on the output trace measured the positive and negative peaks relative to ground.

The Fourier analysis used a $1\,\mathrm{kHz}$ fundamental, nine harmonics and a $2\,\mathrm{s}$ stop time to allow the coupling capacitors to settle. Distortion was calculated from harmonic amplitudes V_n measured using the same convention:

$$\mathrm{THD}_{2:9} = 100\,\frac{\sqrt{\sum_{n=2}^{9} V_n^2}}{V_1}\,\%. \tag{51}$$

Here V_1 in the equation denotes the output fundamental. A separate small-signal AC sweep measured $|V_{\mathrm{out}}/V_{\mathrm{in}}|$ with a unit AC input. Output amplitude and distortion were assessed from the driven sine-wave test.

D. Class AB Power Stage

1) DC Operating Point: The DC analysis examines the circuit with no audio input. It identifies the transistor bias and the internal output voltage. The design targets are $I_{DQ} = 50\,\mathrm{mA}$ in each branch, $11\,\mathrm{mV}$ across each source resistor, and VOUT close to zero. These references define the intended Class AB resting condition.

2) Transient Response: The transient analysis follows the input and output signals over time. It provides the voltage, current, and average-power measurements reported in the results. The method compares settled signal cycles, allowing the output level and the transition between the two transistors to be examined under the same test conditions.

3) Frequency Response and Signal Quality: Frequency-response evaluation examines how the amplifier handles signals at different audio frequencies. Waveform and harmonic analyses assess changes in signal shape, including the transition between the positive and negative portions. These assessments complement the power measurements; the numerical readings presented here correspond to the documented $10\,\mathrm{kHz}$ test.

VI. Simulated Values and Results

A. Preamplification Stage

The preamplifier stage was simulated in NI Multisim to evaluate both the DC bias point and the transient response under real audio signal excitation.

Table VIII compares the calculated theoretical DC parameters with the values obtained from the simulation. The results show close agreement, with maximum percentage deviations below 4.3%. The simulated collector current $I_{CQ} = 1.055$ mA and collector voltage $V_C = 19.466$ V confirm that the 2N2222 transistor operates within its linear active region with sufficient voltage swing headroom. Furthermore, the simulated DC voltage at the output node ($V_{out,DC} = 194.66$ pV) confirms the effectiveness of the output coupling network (C_2) in isolating the DC bias from downstream stages.

Table VIII
COMPARISON OF CALCULATED AND SIMULATED DC OPERATING PARAMETERS

Parameter	Calculated	Simulated	Deviation (%)
Base Voltage (V_B)	3.07 V	3.03 V	1.30%
Emitter Voltage (V_E)	2.24 V	2.33 V	4.02%
Collector Voltage (V_C)	19.65 V	19.47 V	0.92%
Collector Current (I_{CQ})	1.012 mA	1.055 mA	4.25%
Collector-Emitter Voltage (V_{CEQ})	17.41 V	17.14 V	1.55%
Output DC Baseline ($V_{out,DC}$)	0 V	194.66 pV	$\sim 0\%$

Under dynamic excitation, the output audio signal exhibits symmetric voltage excursions reaching approximately $1\ \mathrm{V}_{pp}$ ($\pm 500\ \mathrm{mV}_{\mathrm{peak}}$) centered at 0 V DC, demonstrating clean linear amplification without clipping or crossover distortion.

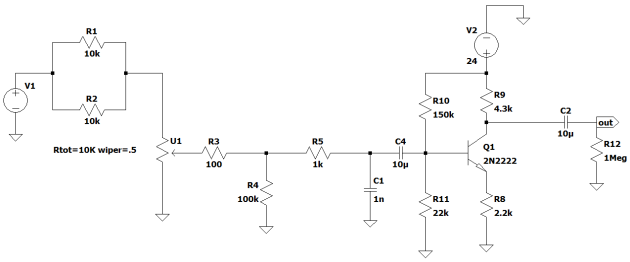


Figure 8. Simulated transient audio output signal V_{out} showing symmetric AC swing centered at 0 V DC.

B. Voltage Amplification Stage

The final circuit delivered $+19.919\,\mathrm{V}$ and $-20.056\,\mathrm{V}$ at the output (Fig. 9). Their difference is $39.975\,\mathrm{V}$ peak to peak, within 0.07% of the $40\,\mathrm{V}$ peak-to-peak target. The resulting gain from the time-domain extrema is

$$A_v = \frac{39.975}{2\sqrt{2}\,(2)} = 7.07\,\mathrm{V/V}. \tag{52}$$

The fundamental corresponds to $20.112/\sqrt{2} = 14.22\,\mathrm{V}$ RMS. Its peak amplitude differs slightly from either oscilloscope cursor because the waveform also contains harmonics.

C. Class AB Power Stage

Tables X and XI retain the available $10\,\mathrm{kHz}$ reference readings for comparison and thermal calculations. The updated schematic defines a single output node shared by the source-resistor junction and SPK2. RMS describes the effective signal level, peak-to-peak describes the full voltage or current swing,

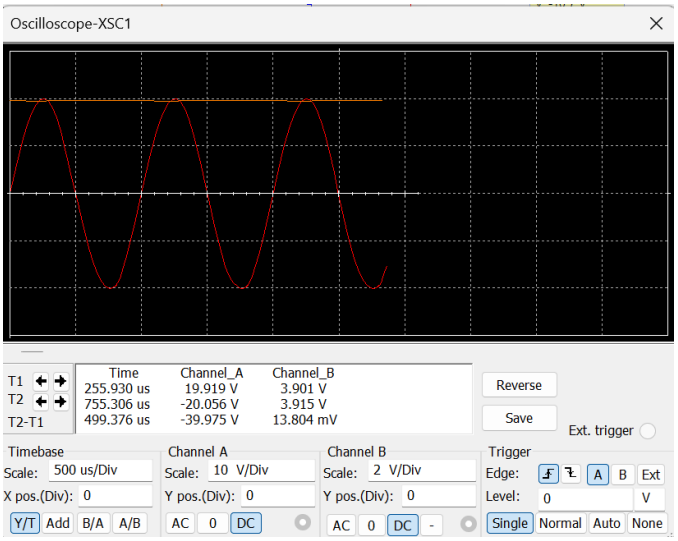


Figure 9. Multisim test at 2 V RMS and 1 kHz. Channel A (red) shows Vout. The cursors mark opposite peaks.

Table IX
FINAL VOLTAGE-STAGE RESULTS WITH THE 43 kΩ LOAD.

Quantity	Simulation result
Internal output at rest (VC4)	−0.125 V
Output DC, Fourier record	0.996 mV
Output fundamental at 1 kHz	20.112 V peak
Second / third harmonic	0.077/0.193 V peak
THD through the ninth harmonic	1.183%
Small-signal gain near 20 Hz	6.930 V/V
Small-signal gain near 1 kHz	7.118 V/V

and mean describes the average DC component. The power readings are average values over the measurement interval.

Table X
RECORDED VOLTAGE REFERENCES AT 10 kHz

Location	Peak-to-peak	RMS	Mean
IRFZ44N gate	39.9 V	14.6 V	+3.62 V
IRF4905 gate	39.8 V	14.6 V	−3.62 V
Output AC reference	38.2 V	13.5 V	—

The output reference is 13.5 V RMS. The dash indicates that a DC measurement for the revised VOUT connection is not included in these reference readings. Its DC adjustment target is 0 V. The gate means describe signal-on operation; I_{DQ} is assessed with the input amplitude set to zero.

Table XI
RECORDED CURRENT AND POWER REFERENCES AT 10 kHz

Quantity	Simulated value
Load-current peak-to-peak	4.78 A
Load-current RMS	1.69 A
Average loudspeaker-load power	22.8 W
IRFZ44N average dissipation	6.58 W
IRF4905 average dissipation	6.63 W
Signal frequency	10.0 kHz

The reference output-power reading exceeds the 20 W design reference. The transistor dissipation values describe heat produced within the output devices and are accounted for separately from the power received by the load.

VII. ANALYSIS AND INTEGRATION

A. Preamplification Stage

The calculated small-signal gain $A_v \approx -2.13$ V/V provides moderate, highly controlled amplification without driving the output into saturation or cut-off under standard audio line-level inputs. By setting the quiescent collector voltage V_C to approximately 9.81 V, the collector node retains ample headroom for positive and negative voltage swings.

When integrated into the complete audio system, the preamplifier's output node connects via C_2 to the input resistor of the TL081-based summer/inverting stage ($R_{11} = 2.2$ kΩ). The output coupling network (C_2 and R_5) successfully eliminates the +9.81 V DC offset from the colector of Q_1, ensuring the TL081 receiving terminal operates centered at 0 V DC. Furthermore, the unbypassed topology eliminates clipping at the peak excursions, matching the downstream headroom requirements of the Class AB output driver stage.

B. Voltage Amplification Stage

The resistor changes were evaluated at comparable output levels. At 1 kHz, one intermediate network produced 15.2 V RMS with 4.87% THD. A lower-resistance network produced 15.1 V RMS with 2.85% THD. In the final network, increasing R_{12} from 3.3 to 3.48 kΩ brought the measured swing from 40.80 to 39.98 V peak to peak and reduced THD from 1.37% to 1.18%. The final choice therefore reflects both the voltage required at the interface and the shape of the signal.

The standalone test establishes the voltage available to drive the next block. The 43 kΩ load draws a fraction of a milliampere, whereas the Class AB stage supplies the loudspeaker current. When the stages are connected, the input bias network of the power stage becomes an additional load. The corresponding interface check is to measure the voltage at the power-stage input under that load.

C. Class AB Power Stage

The recorded reference output voltage supports the average-power result. Applying (23) gives

$$P_L = \frac{(13.5)^2}{8} = 22.78\,\text{W}, \tag{53}$$

which agrees with the $22.8\,\mathrm{W}$ simulation reading after rounding. The expected current is $13.5/8 = 1.69\,\mathrm{A}$ RMS, also consistent with the recorded value.

The $38.2\,\mathrm{V}$ peak-to-peak output corresponds to approximately $19.1\,\mathrm{V}$ peak. Compared with the specified $20\,\mathrm{V}$ peak input, the output amplitude is 95.5% of the input amplitude. This small reduction is consistent with the power stage's role of supplying current while preserving the signal polarity and most of its voltage amplitude.

In the recorded reference test, the delivered power is 14% above the $20\,\mathrm{W}$ objective. This result provides the comparison baseline for the updated power-stage schematic. The two transistor losses are similar, and both remain below the $10\,\mathrm{W}$ per-device thermal design allowance.

For integration with the complete amplifier, the preceding stage provides the required input amplitude while driving the power-stage input. The bias references describe the resting condition, while the reported voltage, current, and power values describe operation with the audio test signal applied.

VIII. Conclusions and Next Steps

A. Preamplification Stage

The discrete 2N2222 BJT preamplifier stage was successfully designed and mathematically verified. By employing an unbypassed emitter topology, the stage achieves stable AC voltage gain while preserving signal linearity and preventing clipping. The DC bias point ($V_{CEQ} = 3.12$ V, $I_{CQ} = 3.02$ mA) ensures linear active operation, with a negligible thermal dissipation of 9.42 mW that keeps operating temperatures near $26.88°$C.

Next Steps:

- Perform transient sweep simulations in LTspice/Multisim to confirm dynamic AC parameters, frequency response, and Total Harmonic Distortion (THD).
- Verify experimental integration with the TL081 mixing stage to adjust overall systemic gains.
- Validate physical breadboard performance against calculated bias points.

B. Voltage Amplification Stage

At $1\,\mathrm{kHz}$ the simulated voltage stage converted a $2\,\mathrm{V}$ RMS input into a $39.975\,\mathrm{V}$ peak-to-peak output across $43\,\mathrm{k\Omega}$. The two measured peaks were within $0.1\,\mathrm{V}$ of the nominal $\pm 20\,\mathrm{V}$ interface level, with 1.18% THD. For the prototype, the first checks are the resting voltage at VC4 and the output swing with the power-stage input connected. Comparing those readings with the standalone results will show the effects of loading and device variation.

C. Class AB Power Stage

The complementary IRFZ44N/IRF4905 stage retains the Class AB topology and uses commercial component values. Its bias design includes a $I_{DQ} = 50\,\mathrm{mA}$ resting-current target, with trimmers for adjustment and source resistors that provide current references. C1 and C2 couple the input signal to the transistor gates, while the load connects directly to VOUT. The thermal assessment covers transistor, resistor, and capacitor losses and gives calculated temperature estimates for the stage electronics.

With a $20\,\mathrm{V}$ peak input at $10\,\mathrm{kHz}$, the recorded reference test delivered $22.8\,\mathrm{W}$ to an $8\,\Omega$ load, exceeding the $20\,\mathrm{W}$ design reference. The output reaches $13.5\,\mathrm{V}$ RMS and $1.69\,\mathrm{A}$ RMS. These reference results support the power-delivery objective for the documented test. The updated DC assessment uses the direct output connection and the $50\,\mathrm{mA}$ adjustment target. The thermal calculation gives MOSFET junction temperatures of $64.5\,^{\circ}\mathrm{C}$ and $59.8\,^{\circ}\mathrm{C}$. Frequency-response, signal-quality, and connected-stage assessments complete the evaluation framework for the amplifier.

References

[1] T. L. Floyd, *Dispositivos electrónicos*, 8th ed. Mexico: Pearson Educación, 2008, secs. 7–2 and 9–2.

[2] International Rectifier, "IRFZ44NPbF: HEXFET power MOSFET," datasheet, document PD-94787B, pp. 1–4.

[3] Infineon Technologies, "IRF4905PbF: HEXFET power MOSFET," datasheet, rev. 2.1, Apr. 8, 2026, pp. 1–4.

[4] ON Semiconductor, "IRF530: TMOS E-FET Power Field Effect Transistor," datasheet, publication IRF530/D, rev. 2, Aug. 2006.

[5] Vishay Siliconix, "IRF9540 Power MOSFET," datasheet, document 91078, rev. C, Aug. 16, 2021. https://www.vishay.com/docs/91078/91078.pdf.